

CSAI 422 – Assignment 4 Report

Advanced Retrieval-Augmented Generation (RAG)

Overview

This project built an end-to-end RAG pipeline using the PopQA dataset with dense retrieval, query expansion, hybrid search, reranking, citation-grounded generation, and self-reflection.

Dataset & Baseline

A reproducible subset of 200 PopQA samples was used. Dense retrieval was implemented with all-MiniLM-L6-v2 embeddings and FAISS indexing as the baseline.

Advanced Retrieval

Query expansion used Groq Llama3 to improve question clarity. BM25 + Dense retrieval were combined using Reciprocal Rank Fusion. Cross-encoder reranking improved final passage ordering.

Grounded Generation

Groq Llama3 generated answers strictly from retrieved passages using citation formatting such as [P1]. Prompting discouraged unsupported claims.

Self-Reflection

A second-pass critique checked grounding, citation quality, and completeness, revising weak answers when necessary.

Results

Dense baseline performed adequately, but hybrid retrieval + reranking significantly improved Recall, Precision, and MRR. The final reflective system produced the best overall answer quality.

Conclusion

The best system combined hybrid retrieval, reranking, grounded prompting, and self-reflection. Retrieval quality was the largest factor in final performance.